

Rahul Kolla

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Summary

Agentic AI Developer and MS Computer Science student specializing in autonomous AI agents, LLM orchestration, RAG systems, and production-grade AI infrastructure. Experienced in building ReAct-based agents, multi-tool workflows, local LLM systems, and backend AI platforms using LangChain, FastAPI, Ollama, OpenAI APIs, and Claude tool use.

Technical Skills

Agentic AI & LLM Systems: AI Agents, ReAct Agents, Multi-Agent Workflows, Tool Calling, RAG Pipelines, Prompt Engineering, Context Management, LangChain, LlamaIndex, OpenAI APIs, Claude APIs, Ollama, Hugging Face, Vector Databases

Programming & Backend: Python, TypeScript, JavaScript, Java, SQL, FastAPI, Flask, REST APIs, Node.js, PostgreSQL, SQLite, Redis

AI/ML: LLM Applications, NLP, Semantic Search, Embeddings, Deep Learning, CNNs, LSTMs

Infrastructure & DevOps: Docker, AWS, Railway, GitHub Actions, CI/CD, Linux, Git/GitHub, Ansible, Webhooks, Structured Logging, Secrets Management

Experience

Software Intern

AiFA Labs – Cerebro SASA Platform

Charlotte, NC
Sep 2025 – Present

- Developed AI-driven workflow automation systems integrating LLM-powered reasoning, backend APIs, and enterprise orchestration pipelines for SAP-based platforms.
- Built and optimized agentic AI workflows involving tool orchestration, prompt engineering, structured data processing, and autonomous task execution in Agile development environments.

Backend Developer Intern

AiFA Labs

Hyderabad, India
Jan 2024 – Jul 2024

- Built scalable backend services and REST APIs using Python and Flask for AI-driven enterprise workflows and real-time data operations.
- Automated ingestion and logging pipelines, improving operational efficiency and supporting production-ready AI platform features.

Projects

Helmsman – Autonomous DevOps AI Agent

Built an autonomous DevOps agent using ReAct-style reasoning loops and Claude tool-use APIs to monitor repository health, dependency risks, CI failures, and PR activity. Integrated GitHub Actions, OSV.dev CVE scanning, and Slack APIs into orchestrated multi-tool workflows with automated issue generation, reporting pipelines, token budgeting, context management, and tool-result caching.

FERRET – Autonomous AI Research Agent

Developed an agentic AI system that autonomously performs web research, synthesizes multi-source information, and generates structured reports with live progress tracking, tool orchestration, and export pipelines.

AI-Powered Smart Document Intelligence System

Engineered a local RAG platform using FastAPI, LangChain, Ollama, and SQLite for semantic document retrieval, summarization, and offline LLM-powered querying.

Automated Security Pipeline for SIEM and IDS

Designed a real-time security monitoring pipeline using Ansible, Wazuh, and Suricata to automate distributed threat detection and infrastructure observability.

AI-Based Tumor Heterogeneity Detection (Patented)

Proposed and implemented an AI-driven approach for quantifying intra-tumor heterogeneity using image-based biomarkers for healthcare diagnostics.

Education

University of North Carolina at Charlotte

Master of Science in Computer Science

Charlotte, NC
2024 – 2026

Vellore Institute of Technology – AP

Bachelor of Technology in Computer Science

Amaravati, India
2020 – 2024

Certifications

Hugging Face Agents Course

Hugging Face, 2025

Microsoft Azure Fundamentals (AZ-900)

Microsoft, 2024

Deep Learning Institute

NVIDIA, 2024

Prompt to Prototype

Google, 2025